

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of M. Sc. (AOC) Examination May 2018

OC 728 Organic Name Reactions and Reagents

Date: 02/05/2018

Day: Wednesday

Time: 01.30 p.m. to 02.00 p.m.

Maximum Marks: 20

MCQ

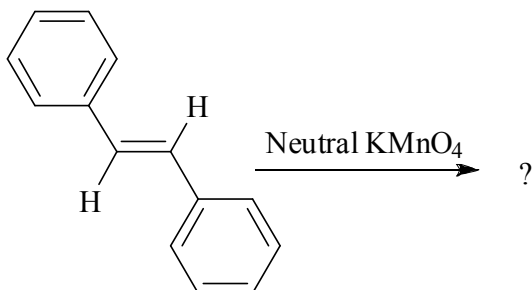
Important Instructions:

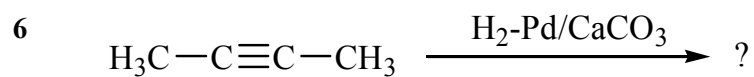
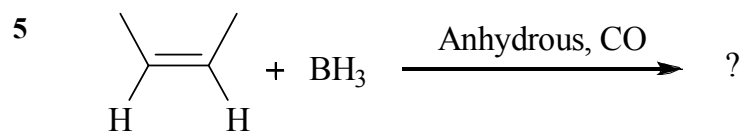
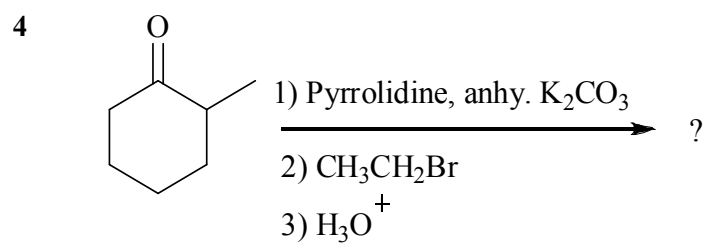
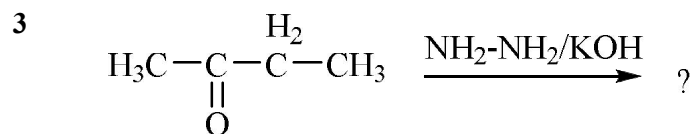
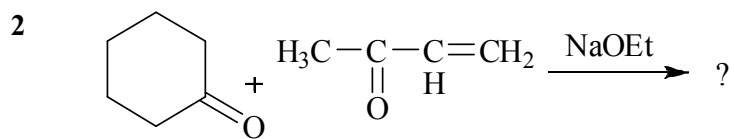
- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

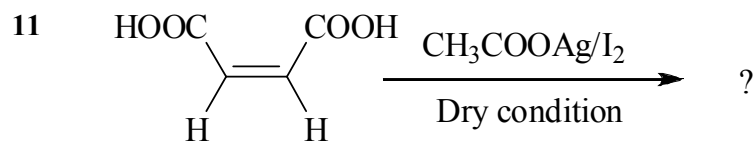
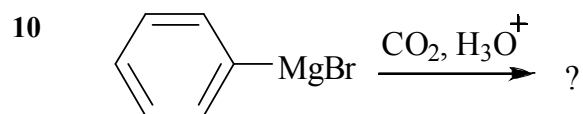
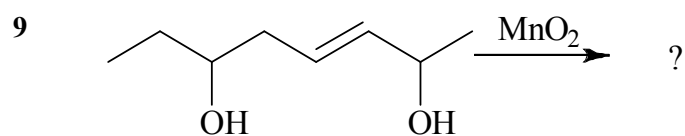
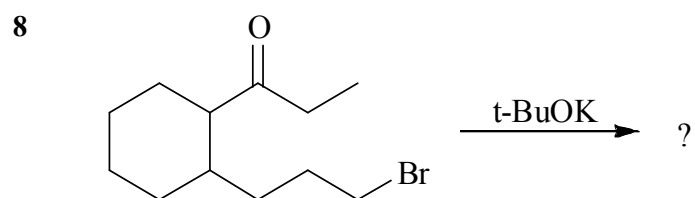
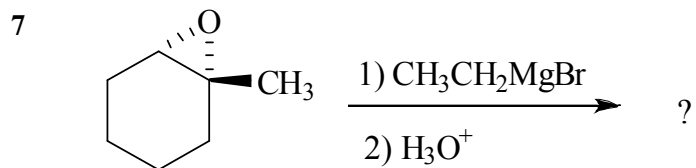
Q-I Draw the structure of final product in following reactions.

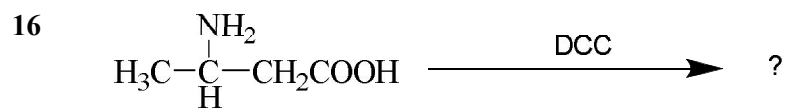
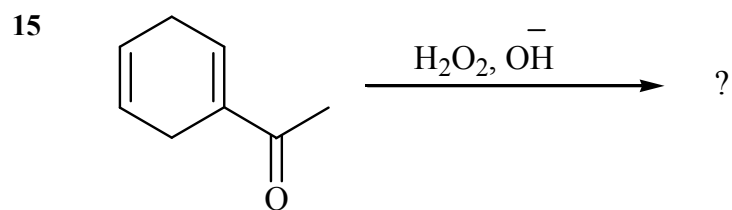
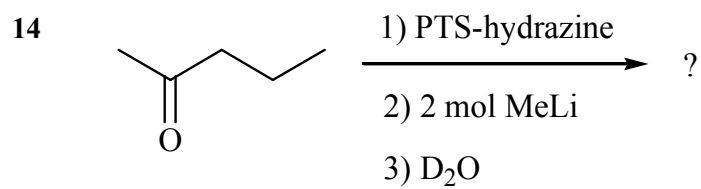
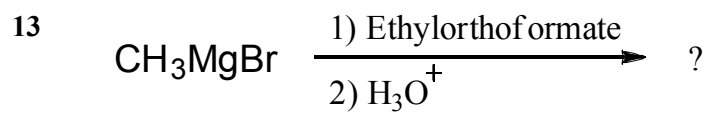
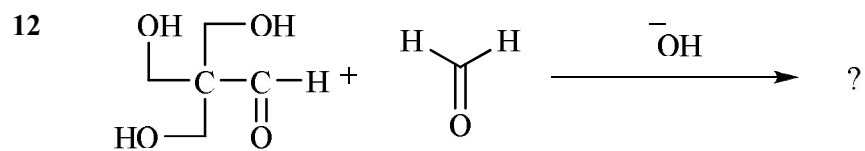
20

1

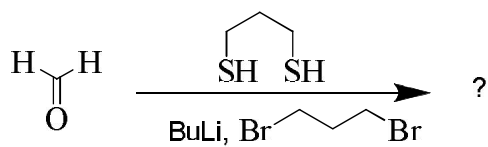




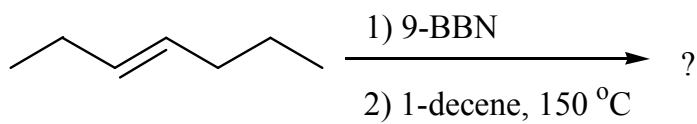




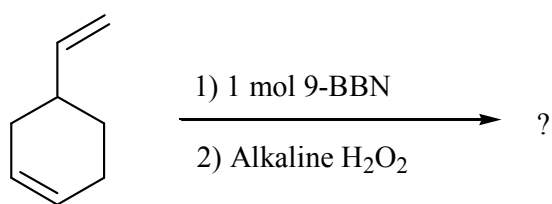
17



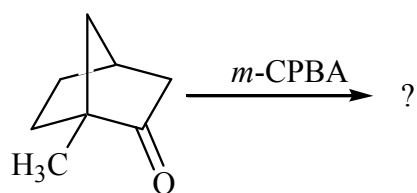
18



19



20



CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of M. Sc. (AOC) Examination May 2018

OC 728 Organic Name Reactions and Reagents

Date: 02/05/2018 Day: Wednesday Time: 02.00 p.m. to 04.30 p.m. Maximum Marks: 50

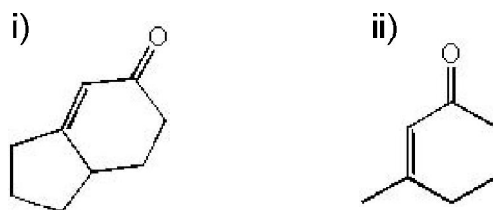
Important Instructions:

- Section I and II must be attempted in TWO ANSWER SHEET.
- Make suitable assumptions and draw neat figures wherever required.
- Use of non-programmable calculator is allowed.
- Show necessary calculations.

SECTION-I

Q-II Answer the following questions as directed.

A Give the synthesis of following compounds via Robinson Annulation. 04



B Discuss the Schlosser modification in detail. 03

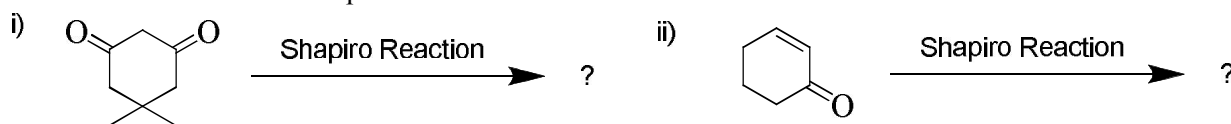
C Explain: Formation of enolate ion using LDA is irreversible. 03

OR

C Discuss the Umpolung approach in dithiane. 03

Q-III Answer the following questions as directed.

A Draw the structure of final product and write the mechanism for it. 04

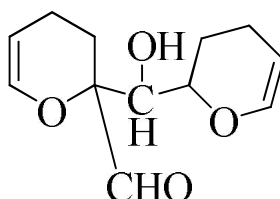


B Give synthesis of disiamylborane and 9-borabicyclo [3.3.1] nonane. 03

C Give synthesis of a bicyclic compound using thexylborane. 03

OR

C Give synthesis of following molecule via Aldol condensation. 03



SECTION-II

Q-IV Answer the following questions as directed.

- A** Discuss: diastereomers will react at different rates with Jones's reagent. **04**
- B** Give synthesis of cyclopentanone using 1,3-dithiane. **03**
- C** Discuss Moffatt oxidation with detail mechanism. **03**

OR

- C** Discuss Kornblum oxidation with detail mechanism. **03**

Q-V Answer the following questions as directed.

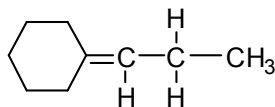
- A** Discuss mechanism for Oppenauer oxidation and Meerwein-Ponndorf-Verley reduction. **04**
- B** Give synthesis of cyclohexanol derivative using di-Grignard reagent. **03**
- C** Give synthesis of γ -lactone using DCC. **03**

OR

- C** Write working mechanism of Wilkinson's catalyst. **03**

Q-VI Answer the following questions as directed.

- A** Explain the formation of combination product via free radical mechanism from aniline using Potassium ferricyanide catalyst. **04**
- B** Write the difference between reactivity of LiAlH_4 and NaBH_4 . **03**
- C** Give synthesis of following compound via Wittig reaction. **03**



OR

- C** Give synthesis of following compound via Wittig reaction. **03**

